BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. According to -------- hypothesis, aerobic bacteria developed into mitochondria:

(A) Symbiotic
(B) Endosymbiont
(C) Mutualistic
(D) Both A and B

Q2. Chromosomal part which uncoils during interphase [is] called:

(A) Euchromatin
(B) Heterochromatin
(C) Chromatin
(D) Both A and B

Q3. Parthenocarpy is artificially induced by :

(A) Cytokinins
(B) Auxin
(C) Ethene
(D) Abscisic acid

Q4. The position of a gene on the chromosome is called its :

(A) Locus
(B) Genotype
(C) Phenotype
(D) All of these

Q5. Pairing of homologous chromosomes is called :

(A) Bivalent
(B) Tetrad
(C) Synapsis
(D) Crossing over

Q6. The membrane that bounds vacuole is called:

(A) Tonoplast
(B) Symplast
(C) Apoplast
(D) All of these

Q7. What is our principle source of energy :

(A) Nuclear energy
(B) Geothermal energy
(C) Solar energy
(D) Tidal energy

Q8. A plant has a growth pattern called:

(A) Closed growth
(B) Open growth
(C) Primary growth
(D) Secondary growth

Q9. During PCR thermostable enzyme is used named as :

(A) DNA polymerase
(B) Taq polymerase
(C) Both A and B
(D) None of these

Q10. Bats use ------ for evaporative cooling :

(A) Saliva
(B) Urine
(C) Shivering
(D) Both A and B

Q11. Autosomal non-disjunction may occur in other than :

(A) 20th chromosome
(B) 21st chromosome
(C) 23rd chromosome
(D) None of these

Q12. Sarcoplasmic reticulum surround each :

(A) Myofilament
(B) Myofibril
(C) Sarcomere
(D) Both A and B

Q13. The actual location of place where an organism lives is called :

(A) Biosphere
(B) Lithosphere
(C) Atmosphere
(D) Habitat

Q14. pBR 322 would enable separating out colonies of bacteria in a medium containing :

(A) Tetracycline
(B) Ampicillin
(C) Gel
(D) Both A and B

Q15. Fresh water ecosystem covers less than :

(A) 10%
(B) 05%
(C) 02%
(D) 01%

Q16. The epiblast is presumptive :

(A) Ectoderm
(B) Mesoderm
(C) Endoderm
(D) Both A and B

Q17. The normal speed of nerve impulse in human is -------- per second :

(A) 100 m/sec
(B) 120 m/sec
(C) 150 m/sec
(D) None of these

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) How animals of hypotonic environment osmoregulate? Give examples.

(ii) Animals excrete nitrogenous wastes with digestive feces. Give example and significance of this adaptation.

(iii) How land animals trap a thick layer of air around the body? Give its significance.

(iv) What is Ecdysis?

(v) Differentiate Hinge Joint and Ball and Socket joint by giving example.

(vi) What is arthritis?

(vii) Define seed dormancy. Give its significance.

(viii) Can we find a fruit without seeds? Give example.

(ix) What is eutrophication?

(x) Differentiate between prairies and savanna.

(xi) Define soil. Give its basic constituents.

(xii) What are industrial effluents? Give their two effects.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What happens when an impulse reaches a synaptic knob?

(ii) Elaborate habituation as simplest form of learning.

(iii) Give negative effects of nicotine.

(iv) What is a sex limited trait?

(v) State sexual dimorphism in drosophila.

(vi) Define linkage group.

(vii) How gene therapy helps cancer patients?

(viii) What are molecular scissors? How were they obtained?

(ix) Write down the role of lambda phages as a vector.

(x) Interpret the role of decomposers in recycling.

(xi) Compare hydrosere with that of xerosere.

(xii) What is parasitism? Write down its importance.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Compare gastrulation and organogenesis.

(ii) How inhibitory effect and compensatory effect are caused?

(iii) What is Karyotype? Give its application in species recognition.

(iv) Give the composition of chromosomes.

(v) Differentiate between heterochromatin and euchromatin.

(vi) What is mitotic apparatus?

(vii) How cancer cells can be distinguished from normal cells?

(viii) What is modern synthesis or Neo-Darwinism?

(ix) What are analogous organs? Give example.

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5. (Marks 4) (a) Discuss the temperature classification of animals.

(b) Discuss nitrogen depletion and its remedies. 4

Q6. (Marks 4) (a) Compare sclerenchyma cells with collenchyma cells.

(b) What is transcription? How it is carried out in cell? 4

Q7. (Marks 4) (a) Explain the steps of that mechanism which maintains the concentration of secretions in the body.

(b) Write a note on ozone depletion. 4

Q8. (Marks 4) (a) Describe the phenomena of fruit set and fruit ripening.

(b) Explain the process of crossing over with the help of diagram. 4

Q9. (Marks 4) (a) Define and explain embryonic induction.

(b) p + q = 1. Argue that this balance shown in theorem may not vary for a non-evolving population? 4

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. Bivalents or tetrads are formed in :

(A) Zygotene (B) Leptotene

(C) Pachytene (D) Diakinesis

Q2. In microcephaly, the individuals are born with small :

(A) Eyes (B) DNA

(C) mRNA (D) tRNA

Q3. A powerful tool of forensic science is a test:

(A) RNA (B) DNA

(C) mRNA (D) tRNA

Q4. The structures that lack secondary wall are :

(A) Fibers (B) Sclerenchyma

(C) Parenchyma (D) Collenchyma

Q5. Treasure of all type of resources is :

(A) Weather (B) Climate

(C) Environment (D) Water

Q6. Particular array of chromosomes that an individual possesses is called :

(A) Holotype
(B) Karyotype
(C) Neotype
(D) Paratype

Q7. Hormone that suppresses ovulation is :

(A) Testosterone
(B) Oestrogen
(C) Progesterone
(D) Gastrin

Q8. The most concentrated environment is termed as :

(A) Hypertonic
(B) Isotonic
(C) Hypotonic
(D) Peritonic

Q9. Number of linkage groups in man is :

(A) 20
(B) 22
(C) 21
(D) 23

Q10. The study of single population's relationship to environment is called :

(A) Autecology
(B) Synecology
(C) Ecology
(D) Gerantology [sic, print for 'Gerontology']

Q11. The most critical phase of mitosis is :

(A) Prophase
(B) Metaphase
(C) Anaphase
(D) Telophase

Q12. The total gestation period (pregnancy) is usually about :

(A) 250 days
(B) 260 days
(C) 270 days
(D) 280 days

Q13. In Sindh, the desert ecosystem is called :

(A) Thal
(B) Sahara
(C) Thar
(D) Ghobi

Q14. Acetabularia is unicellular:

(A) Fungus
(B) Alga
(C) Yeast
(D) Protozoa

Q15. The dissolving cells are called :

(A) Stem cells
(B) Osteoclast
(C) Osteoblast
(D) Osteocytes

Q16. Viral infections can be diagnosed by :

(A) PCR
(B) Cloning
(C) Translation
(D) Transformation

Q17. Among the scientists who believed in divine creation was:

(A) Charles Darwin
(B) Alfred Wallace
(C) Carolus Linnaeus
(D) Jean Lamarck

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What components of internal environment are affected by external fluctuations? How are these corrected?

(ii) What may happen to a cell when placed in a hypotonic environment and then in a hypertonic environment?

(iii) Justify the statement, "Excretion of uric acid in some terrestrial animals is an adaptation to conserve water".

(iv) Compare sapwood and heartwood.

(v) Write the name of unpaired bones of human cranium and face.

(vi) What is osteoporosis? How is it treated?

(vii) How do plants detect light or dark period?

(viii) What is vitamin? How is it produced in plants?

(ix) What do you mean by the productivity of an ecosystem? How is it determined?

(x) What kind of soil conditions are found in grassland ecosystem?

(xi) What is nutrient cycle? How is this cycle disturbed?

(xii) Write the sources and harmful effects of CFCs and SO2.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) How plants respond to stimuli?

(ii) In what way nerve impulse triggers the action potential?

(iii) How different modalities of sensation work?

(iv) What are multiple alleles? Mention their presence in diploid and haploid organisms.

(v) What is test cross? Write down its uses.

(vi) Differentiate between gene linkage and linkage group.

(vii) Write two possible ways to get genes.

(viii) What are plasmids? Give their two examples.

(ix) Write down the two uses of PCR amplification.

(x) Differentiate between autecology and synecology.

(xi) How the trophic levels are involved in the flow of energy?

(xii) What is prey and predator interaction? Write its significance.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) How are area pellucida and area opaca developed?

(ii) The number of older individuals are expected to rise in humans, discuss.

(iii) Why does every genetic code consists of three nucleotides?

(iv) What do you know about the minimal medium used by Beadle and Tatum?

(v) Describe promoter area in transcription.

(vi) Describe mitotic apparatus.

(vii) Describe Turner's syndrome.

(viii) Differentiate between homologous and analogous structures.

(ix) Define Hardy Weinberg Theorem.

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5. (Marks 4) (a) Explain in detail the adaptations in plants to high and low temperature.

(b) What is food web? Give its significance. Draw a food web. 4

Q6. (Marks 4) (a) Explain some major functions of skeletal system in maintenance of human life.

(b) Discuss chemical nature of DNA with reference to nucleoside and nucleotide composition. 4

Q7. (Marks 4) (a) Describe major factors which restore resting membrane potential in a neuron passage of a nerve impulse.

(b) Discuss the importance of forests for human. 4

Q8. (Marks 4) (a) Discuss sexually transmitted diseases. How can these be controlled?

(b) Write an essay on crossing over. 4

Q9. (Marks 4) (a) Discuss abnormal development due to environmental factors and metabolic defects.

(b) Explain natural selection and artificial selection as evidence of evolution. 4